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Design of an Integrated Myocardial Infarction Detection Model using ECG Connectivity Features and Multivariate Time Series Classification

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ABSTRACT Myocardial infarction (MI), commonly known as a heart attack, results from reduced blood flow to a part of the heart. Timely diagnosis of MI is very crucial due to its high mortality rate, especially among older individuals. The existing manual MI diagnosis methods using the electrocardiogram (ECG) signal necessitate the availability of qualified medical professionals while also suffering from human errors and biases. To address this, recently many methods have been proposed to automate MI diagnosis, particularly using machine learning and deep learning. However, most of these methods often employ advanced deep learning architectures like CNN or RNN directly on raw ECG data and hence require considerable computational time and power. In contrast to this, the present paper introduces an innovative MI diagnosis method wherein the multi-lead ECG signal is uniquely modeled as a multivariate time series signal to extract the multivariate sequential features of the signal. These features are then combined with the proposed novel connectivity-based features of ECG signal that exploit the relational information among ECG leads. These combined features, which uniquely encode both the sequential and relational information of the multi-lead ECG data, are then provided to a simple logistic regression classifier for classification, thus reducing the model's computational complexity and time which is extremely important in timely detection of MI. Further, the most informative ECG leads for MI detection are identified to make the model even lighter. The state-of-the-art performance of the proposed integrated model on the PTB-XL dataset verified its efficacy in the MI diagnosis.

INDEX TERMS Myocardial infarction, electrocardiogram, connectivity, multivariate time series classification, dynamic time warping, ROCKET

I. INTRODUCTION

Myocardial infarction (MI), commonly referred to as a heart attack, occurs when there is a decrease or complete cessation of blood flow to a specific region of the heart called the myocardium. Timely detection of MI is of utmost importance, as it remains the leading cause of death among middle-aged and elderly individuals worldwide causing more than 15.2 million deaths worldwide annually, with most of those resulting from the delay in the detection and treatment of MI [1]. Presently, the diagnosis of MI involves the evaluation of the subjects' electrocardiogram (ECG) signals by a skilled medical professional, thus demanding the timely availability of the same and also introducing the potential for human error and observer bias. To address these challenges, in recent years, a lot of

attempts have been made to automate the diagnosis of MI from ECG signals without any human intervention, using various methods ranging from the classical signal processing-based methods, machine learning-based methods to more recent deep learning-based methods [2-11]. In the realm of signal processing and machine learning-based methods, Dohare et al. [2] proposed a model that uses a support vector machine (SVM) classifier and traditional features like the peak-to-peak amplitude, area, mean, etc., and were able to get an accuracy of about 96.66%. Authors in [3] solved the MI detection problem by using a least absolute shrinkage and selection operator (LASSO) regression model and achieved an average accuracy of 95.0%. [4] attempted the automated detection of posterior myocardial infarction from vectorcar-

diagram signals, using the Fourier–Bessel Series Expansion-based Empirical Wavelet Transform. Their approach yielded an accuracy of approximately 95.52%. Sulthana et al. [5] extracted signal features using probabilistic PCA, and multi-linear regression, and classified those using RBF-based SVM classifier to achieve an accuracy of 94.0%. [6] exploited the harmonic phase distribution pattern of ECG data for myocardial infarction identification to obtain an accuracy of 95.6%.

While the classical signal processing and machine learning-based methods perform reasonably well, these methods typically require handcrafted features that often demand a deeper understanding of the input data and its relation with the expected classification outcome. So, in recent years, with the advancement in the field of deep learning, deep learning-based MI detection algorithms witnessed a surge, thanks to their ability to extract abstract features on their own [12]. As far as deep learning-based MI detection is concerned, the majority of the approaches utilize convolutional neural networks (CNNs) or recurrent neural networks (RNNs), which can be observed through a few representative works discussed next. Starting with an interesting work in [7], the authors here proposed a novel deep learning-based MI detection approach using evolutionary multi-branch networks (MBNs) wherein the MBNs were optimized using a genetic algorithm to obtain an overall accuracy of 90.8%. Han et al. in [8] proposed an innovative approach for detecting and localizing MI by employing a multi-lead residual neural network (ML-ResNet) structure and obtained an accuracy of 95.37%. [9] also used residual neural network (ResNet) along with gradient-weighted class activation mapping (Grad-CAM) to achieve an accuracy of 92.7%. Authors in [10] proposed another deep learning-based solution to detect myocardial infarction using multi-scale feature concatenation and obtained an accuracy of 95.76%. [11] applied 2-D CNN and Grad-CAM on the ECG signal for MI detection, achieving an overall accuracy of 96.3%. Although these deep learning-based methods perform well and eliminate the need for manual feature extraction, the associated algorithms typically require significantly higher computational power and time as compared to the traditional signal processing and machine learning-based approaches. While working in a practical resource-constrained scenario, this can pose a serious challenge, as in the case of an acute MI, every second matters a lot.

So, in spite of the numerous efforts dedicated to tackling the challenge of MI detection, there is still scope for improvement, particularly in the feature extraction process, wherein the more informative features can be extracted to improve the performance of the traditional classifiers. In particular, existing methods have largely overlooked the multivariate time series nature of the input ECG signal and the relationship among different leads of ECG, both of which can be potentially exploited for the MI diagnosis. To address these limitations of the current approaches, this paper introduces an innovative method for classifying ECG signals for MI

diagnosis wherein both the multivariate time series nature of the ECG signal and the relational information among its leads are exploited. It uniquely models the multi-lead ECG signal as a multivariate time series signal to leverage multivariate time series features extracted through ROCKET (RandOM Convolutional KERNel Transform) [13], along with novel connectivity-based features such as *ConnDTW*, and correlation for the MI classification using a simple logistic regressor.

In traditional time series classification, methods often rely on specific attributes like shape, frequency, or variance as representations. However, ROCKET's convolutional kernels offer a more efficient alternative, eliminating the need for manual crafting of these representations while effectively capturing a richer set of features. Furthermore, unlike state-of-the-art MI detection methods, this approach is the first to consider the connectivity information among different leads of the ECG for MI detection. After verifying the effectiveness of both the multivariate time series-based approach and the connectivity-based approach independently, the paper integrates both through a feature concatenation to improve the detection performance further. Finally, to enhance the clinical applicability of the proposed MI diagnosis model in a resource-constrained scenario, we identify the most informative ECG leads for the MI diagnosis, thereby effectively reducing the requirement of the diagnosis model, both in terms of signal acquisition hardware and the computing software.

In summary, the main contributions of the paper are as follows.

- Designed a novel multivariate time series-based MI detection model wherein the multi-lead ECG signal is uniquely modeled as a multivariate time series signal to exploit the multivariate sequential information in the ECG signal.
- Designed a novel connectivity-based MI detection model wherein, for the first time, the relational information among different ECG leads is used for MI detection.
- Designed state-of-the-art integrated MI detection model wherein both the multivariate sequential and the relational information in the ECG signal are extracted for the MI diagnosis using a traditional classifier.
- Proposed a novel, robust measure to quantify the relational information between two signals where the signals are specifically the time series signals, rather than any general signals.
- Identified the most informative ECG leads for MI detection, which effectively reduces the required resources for signal acquisition and processing and hence widens the practical applicability of the detection model.

The subsequent sections of the paper are structured as follows: Section II provides essential background and explores related work, establishing the foundation for the proposed approach. In Section III, a comprehensive elucidation of the

proposed MI detection models is presented. Performance evaluation of the proposed models and their comparison with the existing methodologies are expounded in Section IV. Finally, section V offers concluding remarks, summarizing the proposed work and highlighting potential future directions.

II. BASIC PRELIMINARIES AND RELATED WORK

This section comprises the overview of the basic concepts and background required to understand the proposed MI detection models presented in the subsequent sections. The general idea of multivariate time series classification (MTSC) is discussed first, followed by the description of ROCKET, one of the most extensively used MTSC algorithms. Next, some of the existing connectivity measures are discussed followed by the overview of dynamic time warping (DTW), the distance measure that is used by us in the proposed model to construct a novel connectivity-based feature.

A. MULTIVARIATE TIME SERIES CLASSIFICATION

Multivariate time series classification (MTSC) is a specialized task in signal processing and machine learning aimed at categorizing multiple simultaneous sequences of data points. Unlike the general signals and data, the data points here are organized in a chronological sequence, thus encompassing the additional sequential information. Each data point in a multivariate time series is distinguished by a range of distinct features, yielding a wealth of information. The task of classifying such multivariate time series essentially involves labeling a series of observations spanning a specific timeframe wherein each observation is characterized by a multitude of attributes or measurements, resulting in a comprehensive and intricate dataset.

With its applications in diverse fields ranging from healthcare, finance, environmental science, to manufacturing, recent years have witnessed a substantial surge in the advancement of time series classification (TSC) algorithms [14-16]. However, the majority of these progressions have primarily concentrated on univariate TSC, which addresses cases featuring a single time series alongside its corresponding class label [16]. Nevertheless, recently various attempts have been made to extend the existing univariate TSC algorithms for the MTSC [14,16-18]. Apart from these extensions, a few MTSC methods have been proposed exclusively to label the multivariate time series [13,14,19-21]. Although these MTSC methods differ from each other in some aspects, most of those use one of the fundamental approaches like the distance-based approach, feature-based approach, dictionary-based approach, and convolution-based approach. A thorough review of the existing MTSC techniques suggested that the convolution-based ROCKET algorithm is one of the best-performing algorithms in a general multivariate time series classification context and hence was used in our proposed MI detection model. With this general idea of MTSC, we next describe briefly the ROCKET algorithm.

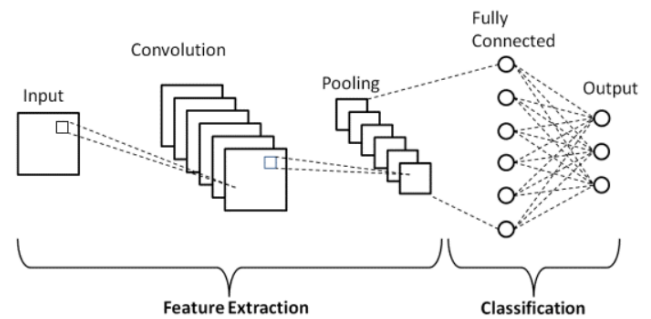


FIGURE 1. Block diagram of the ROCKET multivariate time series classifier [13].

RANDOM CONVOLUTIONAL KERNEL TRANSFORM (ROCKET)

The Random Convolutional Kernel Transform, or ROCKET, was introduced by Dempster and colleagues in 2020 [13]. It leverages an extensive array of 10000 convolution kernels of predefined weights with varying lengths and biases alongside a simple linear classifier like ridge regression or logistic regression. These kernels are individually applied to each instance in the dataset, resulting in a collection of feature maps. From these maps, ROCKET extracts the following two pivotal features:

- 1) Maximum Value: It identifies the highest value within the feature maps, using it as a key classification feature.
- 2) Proportion of Positive Values (ppv): ROCKET introduces a unique feature, ppv, which measures the ratio of positive values within the feature maps.

These two features, maximum value, and ppv, are then employed as inputs to the linear classifier as shown in Fig. 1. This simple approach streamlines and enhances the classification of time series data. ROCKET's efficiency and effectiveness in quick feature extraction using a diverse set of kernels make it an invaluable tool for a wide range of time series classification tasks [13,14]. This allows it to capture a wide spectrum of potential patterns within time series data. As a result, ROCKET established itself as a benchmark in many TSC datasets and gained popularity in various domains like finance, agriculture, and healthcare, where accurate classification of time series data is of paramount importance, thus motivating us to use the same in our proposed MI detection model [13,14,16]. With this essential background of MTSC and the ROCKET algorithm, we next discuss different measures to encode the connectivity or the relational information, that are used in our proposed connectivity-based MI detection model.

B. CONNECTIVITY MEASURES

Connectivity measure between two variables, in general, refers to a metric that quantifies the degree of similarity in the data or signal pertaining to those variables. In the context of multivariate time series data, it essentially refers to a metric

that describes how different variables within the time series are interconnected or how they interact with each other. In this study, initially, we extracted the connectivity information from ECG data using the following four well-established connectivity features and then used those as features for MI classification [22]:

- 1) Coherence: Coherence measures the similarity between the frequency components of two time series signals, evaluating how well the phases and amplitudes of these components correlate across different frequencies.
- 2) Correlation: Correlation assesses the overall linear relationship between two variables, regardless of their frequency, providing a numerical measure of the degree of association between two time series.
- 3) Phase-lag index (PLI): The phase-lag index is a statistical metric used to analyze the degree of phase synchronization or phase coupling between two time series signals. It involves the calculation of the instantaneous phase and the assessment of the stability and consistency of phase differences over time.
- 4) Phase-lag value (PLV): The phase-lag value typically indicates the degree of delay or phase difference between one signal and another at a specific frequency or over a defined time period, quantifying the time delay between the two signals.

Although the aforementioned connectivity measures are crucial for understanding the relationships and interactions among different variables in the general context, some of these are not defined to specifically encode the relational information between the time series data that implicitly has the additional sequential information embedded into it. To exploit this additional sequential information in the multivariate time series data, we proposed a novel connectivity measure using dynamic time warping (DTW), a metric specifically designed to measure the distance between two time series. The proposed DTW-based connectivity measure, termed *ConnDTW*, is then used in our connectivity-based MI detection model to improve its performance further. To appreciate the proposed connectivity measure *ConnDTW* better, a brief overview of DTW is presented next.

C. DYNAMIC TIME WARPING

Dynamic time warping (DTW) is a robust algorithm extensively employed in the realms of time series analysis and data mining. Its primary function is to compare two sequences of data points, accounting for potential variations in their temporal dimension. Unlike conventional distance metrics like the Euclidean distance, which assume a straightforward one-to-one correspondence between data points, DTW introduces a distinctive capability – non-linear alignments. The core objective of DTW is to determine the optimal alignment, often referred to as the warping path, between two sequences while minimizing the overall distance or cost between them. This cost is typically computed based on a chosen distance metric

(commonly, the Euclidean distance) between corresponding data points along the aligned sequences.

For two N-dimensional multivariate time series $\mathbf{X} = [\mathbf{x}_1 | \mathbf{x}_2 | \dots | \mathbf{x}_N]$ and $\mathbf{Y} = [\mathbf{y}_1 | \mathbf{y}_2 | \dots | \mathbf{y}_N]$, the DTW distance between the two, $DTW(\mathbf{X}, \mathbf{Y})$ is defined as:

$$DTW(\mathbf{X}, \mathbf{Y}) = \sum_{k=1}^N DTW(\mathbf{x}_k, \mathbf{y}_k) \quad (1)$$

where $\mathbf{x}_k$ and $\mathbf{y}_k$ denote the individual time series in $\mathbf{X}$ and $\mathbf{Y}$ respectively, while $DTW(\mathbf{x}_k, \mathbf{y}_k)$ indicates the univariate DTW distance between the individual time series $\mathbf{x}_k$ and $\mathbf{y}_k$ [17].

The inherent capability of DTW to accommodate disparities in timing or speed between the two sequences being compared makes it invaluable in applications where even subtle time shifts hold great significance, e.g., speech recognition, and the analysis of biological signals [14,17]. With this motivation, in this paper, we proposed a novel connectivity measure based on DTW, the details of which are discussed in the next section. Now, having equipped with all the necessary background, the following section describes in detail the proposed MI detection models.

III. PROPOSED MI DETECTION MODELS

This section offers a comprehensive discussion of our proposed MI detection models wherein we start with the proposed multivariate time series-based model that exploits the sequential nature of the input ECG signal followed by the connectivity-based model that leverages the relational information within the ECG signal for MI detection. Next, we discuss the proposed integrated MI detection model that integrates the aforementioned models to exploit both the sequential and the relational information from the ECG signal simultaneously to create a more robust and reliable system for MI identification. Finally, we identify the most important subset of ECG leads for MI detection using the proposed integrated model to construct a lighter detection model.

However, to provide a sound basis for our subsequent discussions, we begin with a brief but essential overview of the dataset we used and the preprocessing techniques applied.

A. DATASET

In the present work, we used the PTB-XL ECG dataset, one of the most extensive ECG datasets available, consisting of 21,837 clinical 12-lead ECGs obtained from 18,885 subjects [23]. Each ECG recording in the dataset is 10 seconds in length. These raw waveform data were meticulously annotated by up to two cardiologists, who assigned potentially multiple ECG statements to each record. In total, there are 71 distinct ECG statements that adhere to the SCP-ECG standard, encompassing diagnostic, form, and rhythm statements. Among the population of 18,825 subjects, 52% of the patients are male, while 48% are female. The age range spanned from 0 to 95 years, with a median age of 62 and an interquartile range of 22, demonstrating a wide demographic representation.



FIGURE 2. Block diagram of the proposed multivariate time series-based MI detection model.

B. PREPROCESSING

During the data preprocessing phase, several crucial steps were undertaken to ensure the dataset's quality and suitability for analysis. The first step involved addressing missing values by employing imputation techniques to fill in gaps in the dataset. To eliminate baseline wander, a low-frequency variation often caused by factors like electrode movement, detrending was applied. This involved mathematically removing the baseline wander to focus more precisely on the signal's essential components. Additionally, feature scaling was implemented to standardize all features, preventing any one feature from exerting undue influence based on its magnitude.

Collectively, these preprocessing steps refined the raw dataset, preserving its integrity and establishing a foundation for accurate and meaningful analyses. The dataset had a sampling frequency of 500 Hz, resulting in around 5000 samples per time series with each signal lasting approximately 10 seconds. To increase the dataset size, we utilized windowing techniques to segment each time series into multiple segments, wherein the optimal size of the window length was determined through a comparative performance analysis.

C. PROPOSED MI DETECTION APPROACHES

Following the aforementioned preprocessing steps, three innovative approaches were employed to classify the 12-lead ECG data. These approaches are elucidated in detail below.

1) Multivariate Time Series-based Approach

In this pioneering approach, we took a significant step forward by considering the 12-lead ECG signal as a multivariate time series for the very first time in the MI detection application. This fundamental shift in perspective led to a reimagined strategy for addressing the challenge of MI detection. With this transformation in place, the problem of MI detection essentially evolved as a multivariate time series classification problem, which is then solved using the potent ROCKET algorithm.

The ROCKET algorithm, detailed in the last section, played a central role in our proposed multivariate time series-based MI detection model, particularly in the feature extraction part. The algorithm, with its uniquely defined set of kernels, demonstrated impressive proficiency in abstract feature extraction from the multivariate time series input, specifically the 12-lead ECG signals. These unique multivariate time series-based features, which essentially exploit the sequential nature of the ECG signal, were then applied to different classifiers for the MI classification. Through thorough

comparative analyses of various state-of-the-art classifiers, as presented in the next section, a simple logistic regression classifier was found to be an optimal classifier and hence was used in the proposed MI detection model. The classifier was fine-tuned using stochastic gradient descent for obtaining the optimal weights and biases to minimize the loss function and enhance the accuracy of the predictions. The block diagram of the proposed multivariate time series-based MI detection model is depicted in Fig. 2.

As described earlier in the last section, each kernel in the ROCKET algorithm contributes its own unique pair of features. So, throughout this model design process, we conducted a series of experiments, adjusting the number of kernels to generate different sets of features optimal for the MI detection and found 7000 kernels to be the optimal. Additionally, we explored various window lengths and determined that a 2-second window size yielded the most accurate results. The application of this windowing technique involved breaking down the ECG signal into discrete 2-second intervals, each containing 1000 data samples. Striking a balance between temporal resolution and computational efficiency emerged as a crucial aspect of our methodology. The details of these experimental studies are presented in the next section.

To classify individual patients, we implemented a majority decision strategy. Each subject was categorized based on the label that prevailed across all segments. Specifically, the 12-lead ECG signal was partitioned into 2-second segments, each comprising 1000 samples. The class that predominated across these segments was assigned to the corresponding sample. This strategic approach helped to mitigate potential fluctuations or anomalies within shorter segments, thereby enhancing the reliability of our predictions.

While this multivariate time series-based classification approach indirectly explored the interrelationships among various time series, we aimed to delve even deeper into this relational information. To achieve this, we introduced a connectivity-based MI classifier, which is detailed in the subsequent subsection.

2) Connectivity-based Approach

As discussed earlier, the conventional methods for MI detection [2-11] typically treat each time series within a multivariate ECG time series as an independent entity, thus overlooking the potentially valuable relational information among them. However, as observed in the detection of some other diseases [24-28], the performance of MI detection can also be improved by harnessing this additional relational

Coherece 12 x 12	Correlation 12 x 12
Phase-lag Index 12 x 12	Phase-lag Value 12 x 12

FIGURE 3. Combined connectivity feature matrix.

information among multivariate time series for classification. Specifically, in the case of multi-lead ECG signals, neighboring ECG leads can share meaningful relational information, which can be highly relevant for MI detection. With this in mind, we propose a connectivity-based MI detection model that extracts this relational information among different ECG leads and leverages it for classification.

In the proposed connectivity-based model, we start by extracting connectivity information from the ECG multivariate time series using four widely recognized connectivity measures: coherence, correlation, phase-lag index, and phase-lag value, which were described in the last section. For a 12-lead ECG signal, each of these measures generates a 12×12 connectivity matrix, representing the unique relational information associated with each segment in the multivariate time series. To integrate the information obtained from all four measures, the four connectivity matrices are then concatenated to construct a comprehensive 24×24 matrix of lead connectivity features as shown in Fig. 3.

This combined connectivity matrix exclusively encodes intricate connections and interdependencies among the ECG leads, thus enhancing the data representation by providing a comprehensive view of the sequential relationships within the multivariate time series. To classify this connectivity feature matrix into different classes, we employed various state-of-the-art classifiers and found the convolutional neural network as an optimal one. The same was then used in the proposed connectivity-based MI detection model for the classification as shown in Fig. 4.

To obtain an optimal classification performance of the proposed model, various configurations of the neural network architecture were explored. Our findings revealed that a convolutional neural network with a convolution layer having 3×3 kernels and 32 feature maps, followed by a 2×2 pooling layer, finally connected to a fully connected decision layer yielded the most accurate results. Further improvement in the classification performance was obtained by employing a 3-D CNN architecture, wherein each of the four connectivity matrices was treated as an independent input channel to the CNN. Although the optimal performance of our proposed connectivity-based model was obtained using a deep learning-based CNN classifier, a comparable performance

was obtained using a simpler SVM classifier and logistic regressor, courtesy of the proposed highly discriminating connectivity features. This allows us to bypass the use of computationally expensive neural networks if required, without much compromise in the performance. More importantly, as detailed in the next subsection, our final integrated MI detection model, which integrates connectivity features with sequential features, performs best with a simple logistic regression classifier, thus completely eliminating the need for deep learning architectures.

The classification performance of any classifier, in general, depends on the efficacy of the features provided to it, and the same holds in the case of our proposed connectivity-based MI detection model. As described earlier, our proposed basic connectivity-based MI detection model, used four commonly used connectivity measures, viz., coherence, correlation, phase-lag index, and phase-lag value to construct a combined feature matrix. Although our proposed model with this combined connectivity matrix performed reasonably well, the performance of the model can be improved further by exploring more robust and discriminative features. With this motivation, we next propose, *ConnDTW*, a novel DTW-based connectivity measure to encode the relational information between two time series. Inspired by DTW, a robust distance measure between two time series, we define a new connectivity measure $ConnDTW(\mathbf{x}, \mathbf{y})$ between two time series $\mathbf{x}$ and $\mathbf{y}$, to be a quantity inversely proportional to $DTW(\mathbf{x}, \mathbf{y})$, the DTW distance between $\mathbf{x}$ and $\mathbf{y}$, as follows:

$$ConnDTW(\mathbf{x}, \mathbf{y}) := e^{-DTW(\mathbf{x}, \mathbf{y})/50} \quad (2)$$

Here, the normalizing factor of 50 in the denominator of the exponent is introduced to prevent the connectivity values from becoming too small. Now, as the $DTW(\mathbf{x}, \mathbf{y})$ increases, i.e., the distance or the dissimilarity between $\mathbf{x}$ and $\mathbf{y}$ increases, the value of the proposed connectivity measure $ConnDTW(\mathbf{x}, \mathbf{y})$ decreases, as it should be, and the same holds vice-versa. With the distance measure of DTW being robust to misalignments and stretching effects, our proposed DTW-based similarity measure *ConnDTW* is also expected to be robust towards the same and hence should act as a more robust connectivity feature for the multi-lead ECG signal. To validate this hypothesis in the present context of MI detection, for each multivariate time series segment, we constructed a 12×12 connectivity matrix that essentially encoded the *ConnDTW* among all the time series in that segment. This new connectivity matrix was then applied as an input to a CNN classifier for the MI classification. An exhaustive comparative analysis, as presented in the next section, revealed that, as expected, our newly proposed connectivity measure *ConnDTW*, performed better than the existing connectivity measures and hence validated the applicability of the same. To enhance the classification performance further, we combined *ConnDTW* with the existing connectivity measures, to construct a modified combined connectivity matrix, which was then fed as an input to a classifier for the MI classification. Superior classification performance with

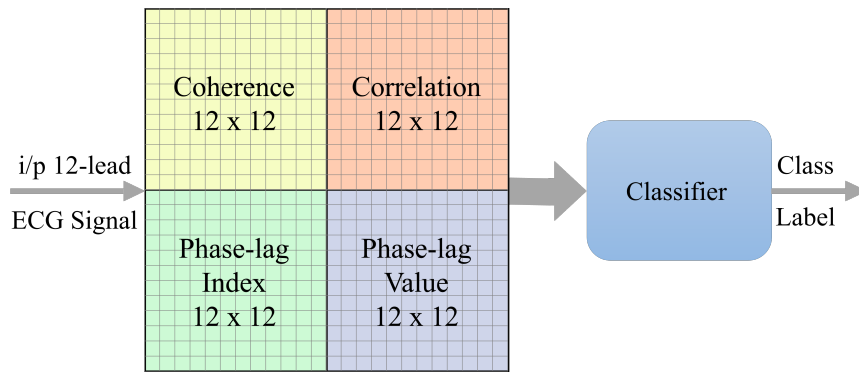


FIGURE 4. Block diagram of the proposed connectivity-based MI detection model.

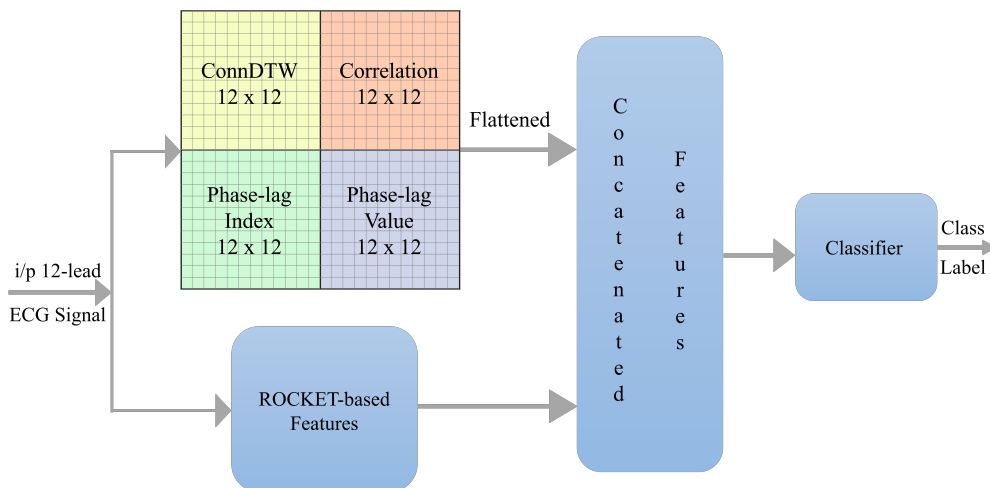


FIGURE 5. Block diagram of the proposed integrated MI detection model.

this modified combined connectivity matrix reassessed the efficacy of ConnDTW in the connectivity-based MI detection approach.

3) Integrated Approach

Having independently validated the effectiveness of both the multivariate time series-based approach and the connectivity-based approach, we next integrate these approaches, envisioning a collaborative framework to elevate the detection performance further. To integrate the two models, we opt for an intermediate-level integration approach [29], wherein the information from both models is fused after the initial model-specific feature extraction, but before the final class decision, as depicted in Fig. 5.

As can be seen from Fig. 5, in the integrated MI detection model, the time series features obtained from the multivariate time series approach, courtesy of the ROCKET algorithm, are concatenated with the combined connectivity features obtained from the connectivity-based approach. This feature fusion results in a singular, enriched feature vector that encompasses both the sequential and the relational information in the multi-lead ECG data. This combined feature vector is then provided as an input to various classifiers for the MI

classification.

Similar to the multivariate time series-based model, here, a simple logistic regression classifier performed best. A thorough comparative analysis of all our proposed models and the existing methods, as presented in the next section, revealed that the proposed integrated MI detection model outperforms our earlier proposed models as well as the existing MI detection methods. This improvement in performance can be attributed to the fact the integrated model harnesses the best attributes of both the multivariate time series-based and connectivity-based models. It not only capitalizes on the intrinsic strengths of each but also combines them to deliver superior performance in the realm of MI detection.

After corroborating the efficacy of the proposed integrated model using a simple logistic regression classifier, we next aim to make our MI detection model even lighter, to make it deployable in a sparser resource scenario. To achieve the same, we try to identify the most important subset of ECG leads that can be used for MI detection without compromising the performance significantly. We start by evaluating our model performance using a single ECG lead, followed by a subset of two ECG leads, and so on. The experimental study with these different subsets of ECG leads, as detailed

in the next section, revealed that with our proposed integrated MI detection model, only a 4-lead ECG setup is sufficient to achieve the MI detection performance that is very close to the original 12 lead ECG setup. This interesting finding essentially manifests the fact that, with better feature extraction, as done in the case of our proposed integrated MI detection model, even the sparser input can be sufficient for the classification.

Having described our proposed MI detection models in this section, the next section presents an elaborate account of the classification performance of these models. It also compares their performance with that of the state-of-the-art MI detection methods. This comprehensive assessment offers valuable insights into the individual merits of the models and areas where they may have limitations or scope for improvement.

IV. EXPERIMENTS AND RESULTS

In this section, we begin by showcasing the classification performance of our three proposed MI detection models, i.e., multivariate time series-based model, connectivity-based model, and integrated model. Following this, we conduct a comparative performance analysis of these proposed models with the established MI detection methods to corroborate the applicability of the proposed MI detection models.

Throughout this study, the classification performance of MI detection methods was assessed on the PTB-XL dataset using three widely used performance metrics: accuracy, sensitivity, and specificity which are defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \quad (3)$$

$$\text{Sensitivity} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (4)$$

$$\text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} \quad (5)$$

where TP (True Positives) corresponds to instances where the model correctly predicts the positive class, TN (True Negatives) corresponds to situations where the model correctly predicts the negative class, FP (False Positives) occurs when the model incorrectly predicts the positive class despite the actual ground truth being negative, and

FN (False Negatives) arises when the model erroneously predicts the negative class when the actual ground truth is positive.

We employed a 5-fold cross-validation approach across all the models to evaluate the classification performance wherein the entire dataset is randomly divided into five parts or folds. Four of these folds are used for training, and the remaining fold is reserved for testing. The model is trained on the training set, and metrics like accuracy, specificity, and sensitivity are computed on the testing set. This process is repeated five times, each time using a different fold as the testing set. The average values for accuracy, specificity, and sensitivity are then calculated. By averaging performance

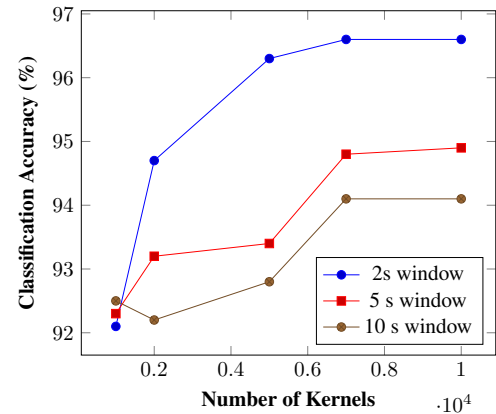


FIGURE 6. Classification accuracy of MI detection for different numbers of kernels and window length.

across multiple folds, we reduce the variability in performance measurements, thus enhancing the reliability of the results.

Starting with the proposed multivariate time series-based model, its performance is expected to depend on the number of kernels in ROCKET, and hence determining its optimal value becomes extremely important. Apart from the number of kernels N , the classification performance of the model is also expected to vary with the length of the input multivariate time series, which in turn is decided by the window length of the ECG segment. So, to find the optimal values of the number of kernels N and the window length w , we constructed models using different combinations of N and w and examined the resulting performance, which is shown in Fig. 6. From these results, it can be observed that 7000 kernels with a window length of 2 seconds, i.e., 1000 samples, perform optimally and hence were selected in the final model.

The results shown in Fig. 6 were obtained using a default classifier in the ROCKET algorithm, i.e., a logistic regression classifier. To obtain the optimal classification performance, we evaluated the performance of the proposed model by using different state-of-the-art classifiers like support vector machine (SVM), random forest, and artificial neural network (ANN); the results of which are shown in Fig. 7. By comparing the results in Fig. 7, it can be concluded that the MI detection model with logistic regression classifier performs the best and hence was used in the proposed model.

As far as the performance of the connectivity-based MI detection model is concerned, it is expected to vary with the choice of connectivity measure, i.e., coherence, correlation, phase-lag index, phase-lag value, and the proposed ConnDTW. Therefore, to determine the most suitable connectivity measure for the MI detection application, we first construct a 12×12 connectivity matrix for each connectivity measure and then apply it as input to a CNN classifier. The resulting classification performance metrics are compared in Fig. 8. Based on the results presented in Fig. 8, it is evident that the proposed connectivity measure ConnDTW stands out as the most effective connectivity measure for MI detection.

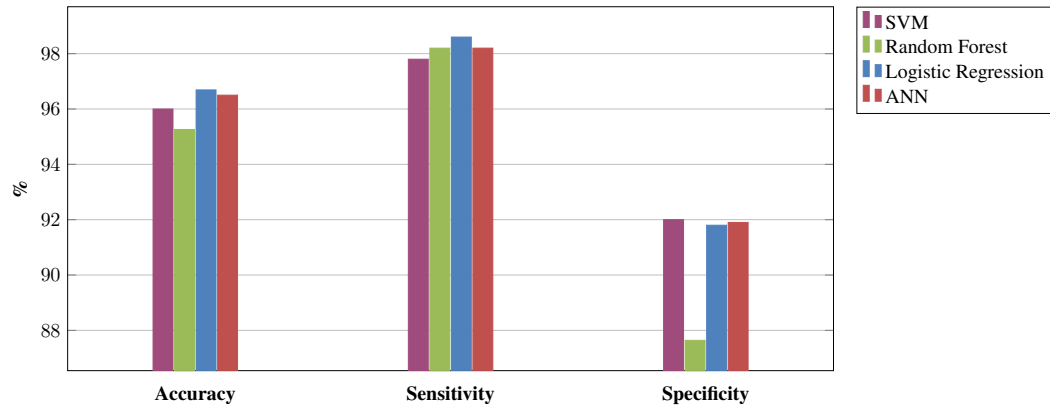


FIGURE 7. Classification performance of the proposed multivariate time series-based MI detection models using different classifiers.

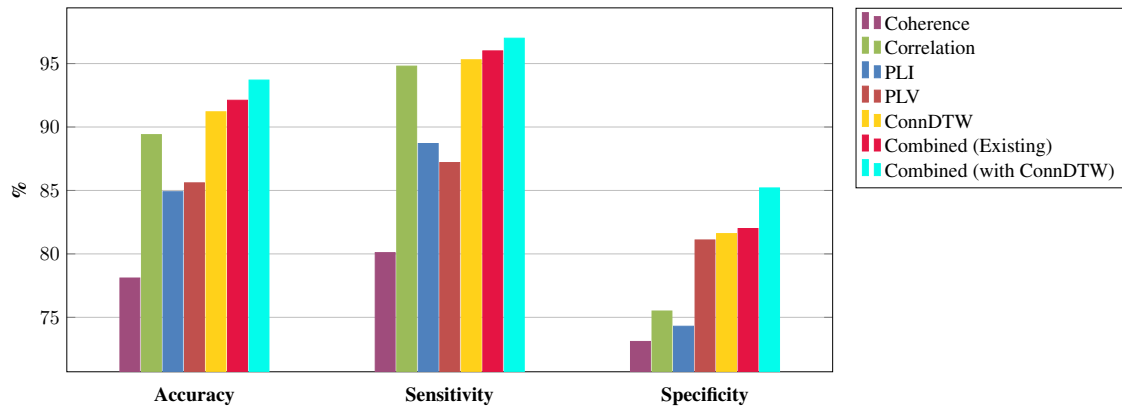


FIGURE 8. Classification performance of the proposed connectivity-based MI detection model using different connectivity measures.

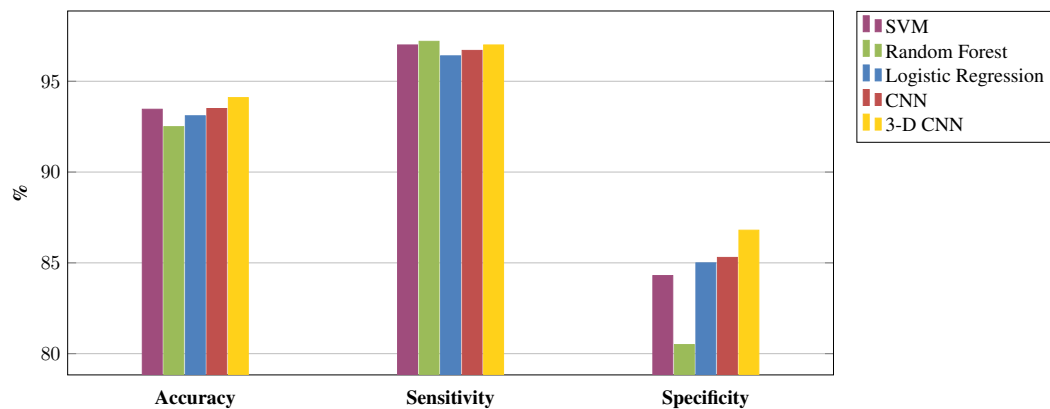


FIGURE 9. Classification performance of the proposed connectivity-based MI detection model using different classifiers.

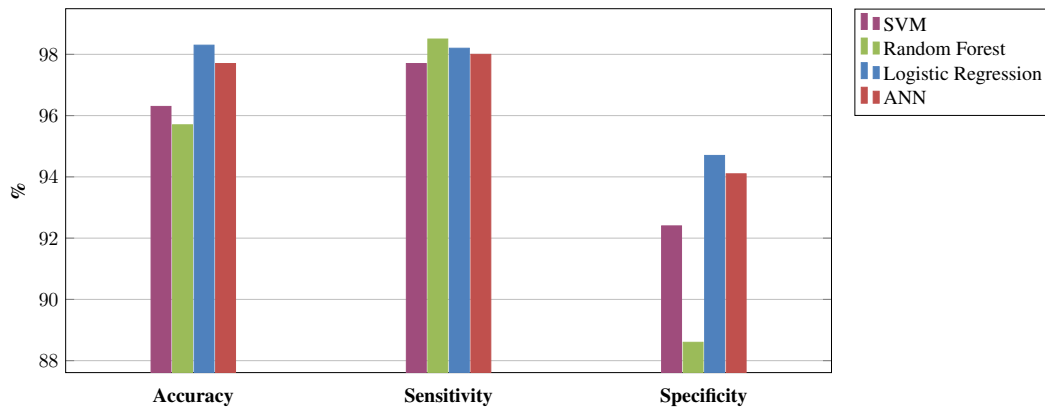


FIGURE 10. Classification performance of the proposed integrated MI detection model using different classifiers.

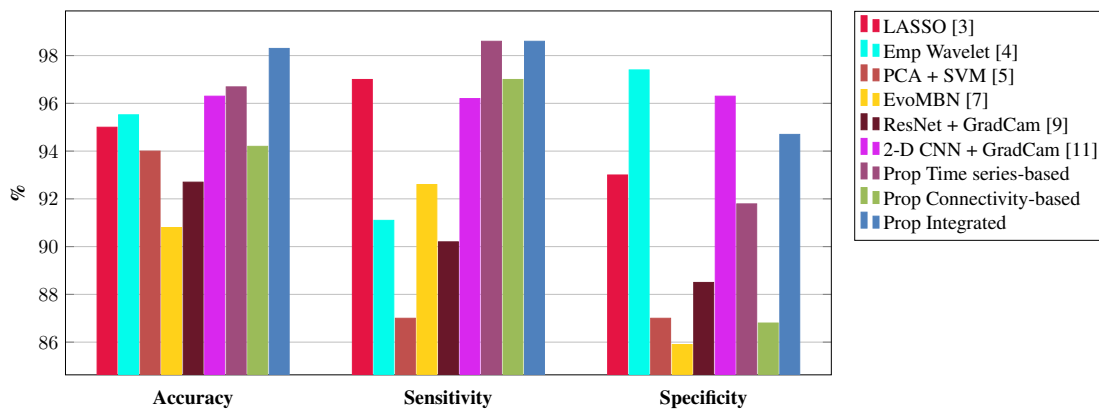


FIGURE 11. Classification performance of different MI detection methods.

Since each connectivity measure carries distinct information, aggregating the data from all of them should enhance the model's performance. To test this hypothesis, we concatenate the four 12×12 connectivity matrices of best-performing measures, i.e., ConnDTW, correlation, phase-lag value, and phase-lag index, to create a comprehensive 24×24 matrix, which is subsequently used as input to the same classifier. The superior performance of the combined connectivity matrix, as demonstrated in Fig. 8, validates our hypothesis. To reassert the advantage of the proposed connectivity measure ConnDTW, we repeated the above exercise, but with the existing four connectivity measures, i.e., coherence, correlation, phase-lag index, and phase-lag value, the results of which are shown in Fig. 8. Comparative analysis of the results presented in Fig. 8 clearly demonstrates the advantage of the proposed connectivity measure in MI detection.

Now, similar to the multivariate time series-based MI detection model, we evaluate the performance of the connectivity-based model using different state-of-the-art classifiers. The performance of the connectivity-based model using the different classifiers is compared in Fig. 9. The results reveal that although the 3-D CNN classifier performs the best, SVM and logistic regression classifiers also produce

comparable results, and hence can be used to make the overall detection model lighter and easily deployable. This can be attributed to the fact that in our proposed connectivity-based MI detection model, the classifiers are already provided with the highly discriminating connectivity features of the ECG signal, rather than the original ECG signal, thus avoiding the need for abstract feature extraction using convolutional neural networks.

After validating the performance of both the multivariate time series-based model and the connectivity-based model, we next evaluate the performance of our integrated MI detection model. Similar to the earlier models, here also we employ various state-of-the-art classifiers for the final classification purpose and compare the resulting performance. From the comparative performance analysis, as shown in Fig. 10, a simple logistic regression classifier turns out to be the optimal one and hence was used in the proposed integrated MI detection model. To corroborate the efficacy of this feature fusion approach, we compare the classification performance of this optimized integrated model with our previously introduced models. The comparison results, as presented in Fig. 11, clearly demonstrate that the integrated model surpasses the performance of our individual models,

TABLE 1. Classification performance of the proposed MI detection model using different ECG leads.

Lead Number	Accuracy	Sensitivity	Specificity
1	0.822	0.861	0.721
2	0.935	0.955	0.888
3	0.839	0.924	0.622
4	0.920	0.963	0.780
5	0.823	0.915	0.607
6	0.937	0.982	0.821
7	0.823	0.927	0.540
8	0.716	0.812	0.498
9	0.745	0.790	0.642
10	0.814	0.857	0.697
11	0.906	0.972	0.742
12	0.954	0.979	0.891

TABLE 2. Classification performance of the proposed MI detection model using different subsets of ECG leads.

Lead Numbers	Accuracy	Sensitivity	Specificity
12	0.954	0.979	0.891
12,2	0.951	0.954	0.943
12,2,6	0.965	0.962	0.948
12,2,6,4	0.970	0.981	0.940

thereby affirming its efficacy.

Having compared the classification performance of our proposed MI detection models, we then compare their performance with that of the state-of-the-art MI detection methods. The comparison results, shown in Fig. 11, clearly indicate a significant improvement in the overall classification performance with the proposed integrated MI detection model and thus attest to the superiority of the same. This improvement in the performance can be ascribed to the fact that, unlike the existing methods, the proposed integrated model uniquely leverages both the sequential and relational information within the ECG data simultaneously.

Finally, to make our proposed MI detection model even lighter, we try to find the optimal subset of ECG leads for which the classification performance is comparable to an original 12-lead ECG setup. To do the same, we first evaluate the classification performance of the proposed model using a single ECG lead signal for different leads and then compare those as shown in table 1. These results reveal that the lead numbers 12, 6, 2, and 4 are most informative and important as far as MI detection is concerned. Having obtained the most informative ECG leads, we next evaluate the model performance using a subset of best two leads, best three leads, and so on. The obtained results for different subsets of ECG leads are tabulated in table 2. By observing these results, it can be concluded that even a small subset of only four ECG leads, with lead numbers 12, 6, 2, and 4 is sufficient to produce a performance that is very close to the one obtained using all 12 ECG leads. This insight is particularly significant in scenarios where there are practical constraints on computational and medical resources. Specifically, in such scenarios, relying solely on these four leads can yield precise and dependable outcomes using the proposed integrated MI

detection model. This reduction in the ECG signal acquisition and processing requirement could be possible due to our novel integrated model that uniquely extracts the highly discriminating sequential and relational features from the available input data, thus bringing down the essential input data requirement.

V. CONCLUSION AND FUTURE SCOPE

The present paper addresses an important medical challenge of timely detection of MI using ECG signals without any human intervention. To solve this problem, three novel MI detection models have been proposed, viz., multivariate time series-based model, connectivity-based model, and integrated model. The multivariate time series-based model uniquely represented the input ECG signal as a multivariate time series signal to exploit the multivariate sequential nature of the input. This novel representation translated the MI detection problem into a multivariate time series classification task, which was then solved using ROCKET-based features and a simple logistic regression classifier. The connectivity-based model, on the other hand, is the first method, to leverage the relational information among different ECG leads for MI detection, wherein the relational information in the ECG signal was encoded using different connectivity measures. To enhance the performance of the connectivity-based model, a novel DTW-based connectivity measure was proposed which robustly quantified the relational information between two time series. Ultimately, both the proposed models were integrated to design an innovative integrated MI detection model, effectively leveraging both sequential and relational information from the ECG signal. The application of the integrated model on the PTB-XL dataset demonstrated state-of-the-art detection performance, achieving superior results with reduced computational complexity. To widen the practicality of the proposed model further, the most informative ECG leads were discovered, thus making the model even lighter.

While the proposed integrated model demonstrated state-of-the-art detection performance, there is still a potential for further improvement by designing or exploring alternative multivariate time series classifiers. The proposed connectivity measure ConnDTW, although proved to be useful in the present MI detection application, can be explored in the general time series applications, to validate its generalizability. Similarly, the application of the proposed integrated approach in problems like Alzheimer's disease detection and schizophrenia detection, involving similar time series data, can form an interesting extension of the present work.

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